# GLOBAL SOLAR ATLAS
BY WORLD BANK GROUP

## Lucknow

26.857530°, 080.912659°
unnamed road, Lucknow, Uttar Pradesh, India
Time zone: UTC+05:30, Asia/Kolkata [IST]

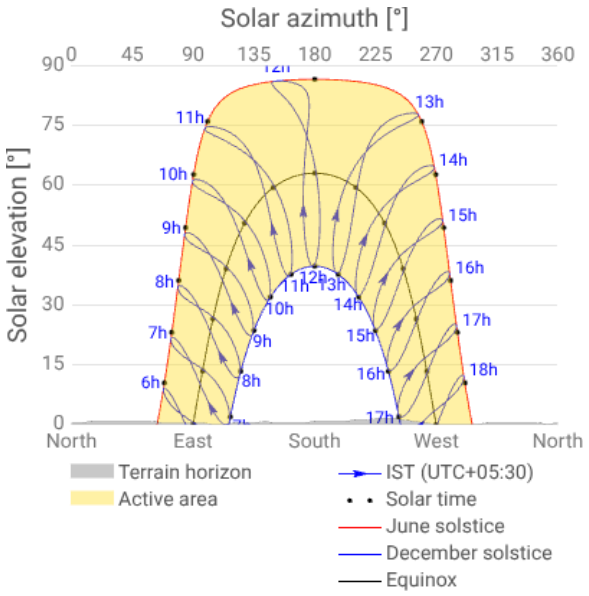
Report generated: 18 Jul 2026

### SITE INFO

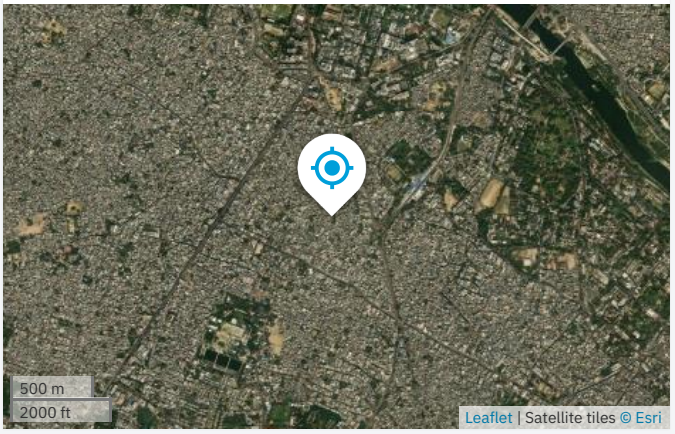
#### Map data

| | | | Per year |
|---|---|---|---|
| Specific photovoltaic power output | PVOUT specific | **1457.1** | kWh/kWp |
| Direct normal irradiation | DNI | **1145.5** | kWh/m$^2$ |
| Global horizontal irradiation | GHI | **1725.6** | kWh/m$^2$ |
| Diffuse horizontal irradiation | DIF | **925.9** | kWh/m$^2$ |
| Global tilted irradiation at optimum angle | GTI opta | **1851.5** | kWh/m$^2$ |
| Optimum tilt of PV modules | OPTA | **24 / 180** | ° |
| Air temperature | TEMP | **24.9** | °C |
| Terrain elevation | ELE | **127** | m |

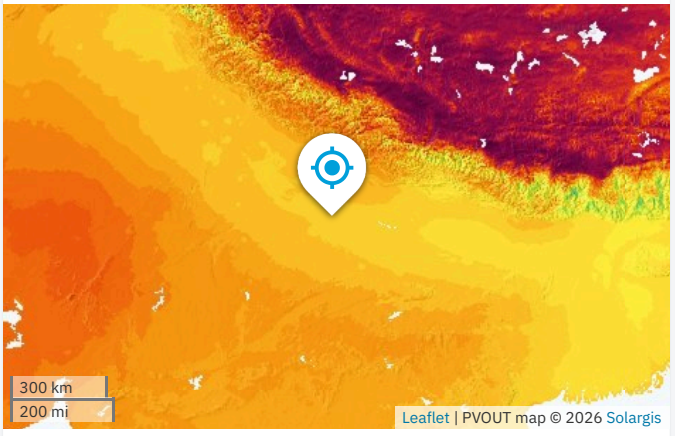
#### Horizon and sunpath




#### Map




#### PVOUT map




PVOUT: Long-term average of annual totals of PV power potential





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## PV ELECTRICITY AND SOLAR RADIATION

### Annual averages

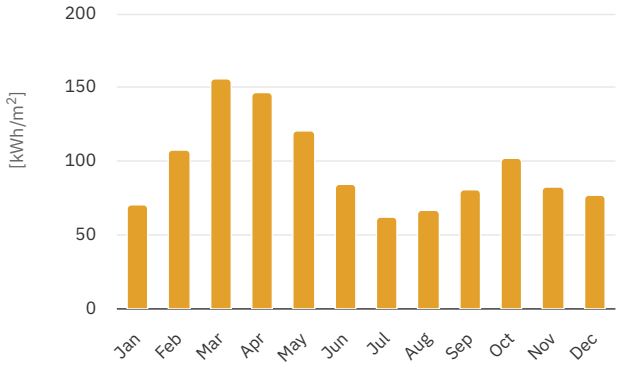
Direct normal irradiation

1154.6

kWh/m$^2$ per year

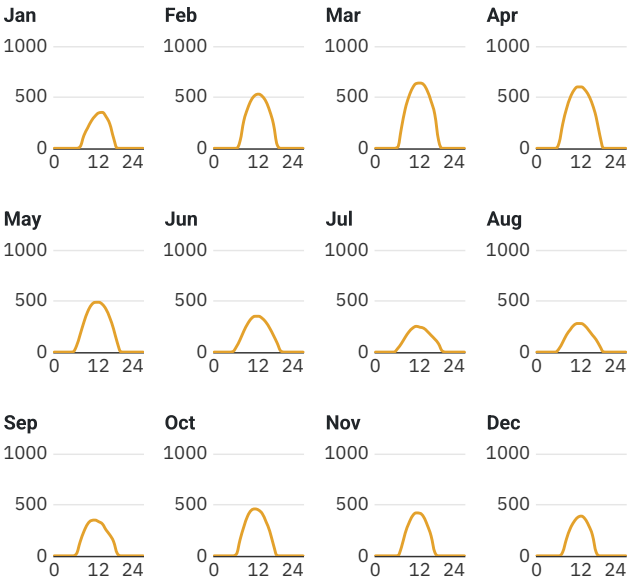
### Monthly averages

Direct normal irradiation




### Average hourly profiles

Direct normal irradiation [Wh/m$^2$]




UTC+05:30

### Average hourly profiles

Direct normal irradiation [Wh/m$^2$]

| | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 - 1 | | | | | | | | | | | | |
| 1 - 2 | | | | | | | | | | | | |
| 2 - 3 | | | | | | | | | | | | |
| 3 - 4 | | | | | | | | | | | | |
| 4 - 5 | | | | | | | | | | | | |
| 5 - 6 | | | | | 1 | 4 | | | | | | |
| 6 - 7 | | | 12 | 53 | 59 | 52 | 36 | 33 | 20 | 17 | 1 | |
| 7 - 8 | 17 | 73 | 198 | 240 | 172 | 122 | 84 | 97 | 137 | 159 | 84 | 39 |
| 8 - 9 | 125 | 268 | 382 | 386 | 291 | 205 | 144 | 168 | 237 | 283 | 220 | 165 |
| 9 - 10 | 198 | 385 | 504 | 495 | 393 | 275 | 197 | 218 | 311 | 387 | 318 | 265 |
| 10 - 11 | 266 | 471 | 590 | 567 | 459 | 326 | 234 | 263 | 345 | 444 | 391 | 332 |
| 11 - 12 | 314 | 517 | 634 | 600 | 486 | 350 | 252 | 279 | 350 | 459 | 420 | 375 |
| 12 - 13 | 346 | 529 | 638 | 600 | 487 | 350 | 243 | 279 | 334 | 447 | 416 | 388 |
| 13 - 14 | 351 | 508 | 623 | 571 | 466 | 329 | 234 | 256 | 312 | 411 | 380 | 361 |
| 14 - 15 | 308 | 456 | 560 | 516 | 417 | 290 | 202 | 214 | 253 | 333 | 298 | 296 |
| 15 - 16 | 237 | 364 | 460 | 429 | 335 | 233 | 164 | 166 | 205 | 241 | 195 | 204 |
| 16 - 17 | 102 | 237 | 331 | 305 | 227 | 163 | 127 | 121 | 141 | 107 | 35 | 36 |
| 17 - 18 | 1 | 34 | 81 | 127 | 94 | 88 | 82 | 58 | 24 | 2 | | |
| 18 - 19 | | | | | 2 | 10 | 11 | 2 | | | | |
| 19 - 20 | | | | | | | | | | | | |
| 20 - 21 | | | | | | | | | | | | |
| 21 - 22 | | | | | | | | | | | | |
| 22 - 23 | | | | | | | | | | | | |
| 23 - 24 | | | | | | | | | | | | |
| Sum | 2,266 | 3,844 | 5,013 | 4,889 | 3,889 | 2,797 | 2,008 | 2,155 | 2,668 | 3,289 | 2,758 | 2,462 |


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## GLOSSARY

| Acronym | Full name | Unit | Type of use |
|---|---|---|---|
| DIF | Diffuse horizontal irradiation | kWh/m², MJ/m² | Average yearly, monthly or daily sum of diffuse horizontal irradiation (© 2025 Solargis) |
| DNI | Direct normal irradiation | kWh/m², MJ/m² | Average yearly, monthly or daily sum of direct normal irradiation (© 2025 Solargis) |
| ELE | Terrain elevation | m, ft | Elevation of terrain surface above/below sea level, processed and integrated from SRTM-3 data and related data products (SRTM v4.1 © 2004 - 2025, CGIAR-CSI) |
| GHI | Global horizontal irradiation | kWh/m², MJ/m² | Average annual, monthly or daily sum of global horizontal irradiation (© 2025 Solargis) |
| GTI | Global tilted irradiation | kWh/m², MJ/m² | Average annual, monthly or daily sum of global tilted irradiation (© 2025 Solargis) |
| GTI_opta | Global tilted irradiation at optimum angle | kWh/m², MJ/m² | Average annual, monthly or daily sum of global tilted irradiation for PV modules fix-mounted at optimum angle (© 2025 Solargis) |
| OPTA | Optimum tilt of PV modules | ° | Optimum tilt of fix-mounted PV modules facing towards Equator set for maximizing GTI input (© 2025 Solargis) |
| PVOUT_total | Total photovoltaic power output | kWh, MWh, GWh | Yearly and monthly average values of photovoltaic electricity (AC) delivered by the total installed capacity of a PV system (© 2025 Solargis) |
| PVOUT_specific | Specific photovoltaic power output | kWh/kWp | Yearly and monthly average values of photovoltaic electricity (AC) delivered by a PV system and normalized to 1 kWp of installed capacity (© 2025 Solargis) |
| TEMP | Air temperature | °C, °F | Average yearly, monthly and daily air temperature at 2 m above ground. Calculated from outputs of ERA5 model (© 2025 ECMWF, post-processed by Solargis) |

## ABOUT

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